



Film-Depth-Dependent Charge Mobilities in Organic Semiconductor Films

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ABSTRACT

Carrier mobility is a crucial parameter characterizing the performance of organic semiconductors in thin film optoelectronic devices. The organic semiconductor film is usually not uniform along the film-perpendicular direction. This non-uniformity affects the carrier mobility at various depths within the film, thereby influencing device performance. Conventional methods for measuring mobility are basically not available to determine film-depth-dependent charge density or mobility. In this study, an easily accessed approach was proposed to investigate the film-depth-dependent electronic properties. Soft plasma etching was introduced to achieve incremental etching of the film, without causing damage to the underlying sublayers. For doped films with intrinsic charge, film-depth-dependent charge density, and film-depth-dependent mobility were extracted from the transfer characteristics evolution during the etching. For materials without intrinsic charge, film-depth-dependences of mobility could also be estimated under light illumination. The proposed method achieves a film-depth resolution of approximately 10 nm. It is revealed that for some model conjugated polymers, the mobility at the top surface is one order higher than that at the buried region of the transistor. This feature leads to unfavorable field-effect mobility. This work provides insights into the film-depth-dependent electronic properties and a strategy to optimize the electronic device performance.

1. Introduction

Organic semiconductor has the advantages of flexibility, solution processing, and adjustable molecular structure, which exhibits highly promising applications in organic optoelectronics [1–3]. The conjugated backbone of organic semiconductors forms overlapping π -orbitals [4]. And the delocalization of π -electrons results in enhanced charge transport properties [5]. Mobility is a crucial electronic property of organic electronic devices including light emission diodes, field-effect transistors (FET) and photovoltaics, which is the derivative of carrier drift velocity with respect to electric field [6–10]. High mobility is often required for achieving high-performance and high-frequency electronics [11]. Therefore, carrier mobility serves as a key performance standard of semiconductors [6,12,13].

The carrier mobility of the semiconductor is affected by many factors such as molecular orientation, crystallinity, doping and temperature [6,14,15]. During the organic film deposition from the solution, the phase evolutions along film-depth directions are not synchronic, which

generally leads to film-depth-dependent morphological variations. Due to the influence of surface and interface properties, polymer molecular weight, solubility, crystallization, solution properties, etc., the film is uneven along the vertical direction. The inhomogeneity includes spatial variations of molecular packing orientation, aggregation, crystallization, etc. (Fig. 1a), and phase separation also occurs in the multi-component systems [16]. Components with low surface energy tend to be distributed near the air interface, while the others are distributed in the lower part of the film [17]. Moreover, uneven doping treatment has been proven to enhance the performance of the semiconductor film. Doping on the surface of organic single crystals was confirmed to enhance the transconductance [18]. It probably causes different transport characteristics of carriers at different locations along the perpendicular direction.

In the classical theory of organic field effect transistor (OFET), only the first few molecular layers near the dielectric interface are involved in charge transport [19,20]. In recent years, there have been reports on the charge transport properties at locations far from the interface within the

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active layer of OFETs. According to Li et al. [21], the actual charge injection in anisotropic organic crystals or highly oriented polymer systems is far away from the channel. This is attributed to the significantly faster charge transport along the channel direction compared to the charge transport from the source to the dielectric/semiconductor interface. As a result, a fraction of charges even reaches the drain before reaching the channel. Therefore, it becomes imperative to investigate the bulk transport of charges in highly ordered systems [22]. It was reported that the mobility of the carriers in the OFET based on 2,20-(2,20-bithiophene-5,50-diyl)bis(5-butyl-5H-thieno[2,3-c]pyrrole-4,6)-dione correlates with the out-of-plane domain size as well as crystallite orientation along the vertical direction. The polycrystalline films with a thickness of 75 nm exhibit superior electrical characteristics, which is associated with the molecular orientational order within the bulk film [23]. The charge transport in F₂-TCNQ (2,5-Difluoro-7,7,8,8-tetracyanoquinodimethane) was explored [24]. All the molecules in F₂-TCNQ single crystal are parallel to each other and display a 3D slipped π -stacking motif, forming a 3D percolation network. Once one path is blocked, carriers can travel along other paths. The 3D mobility was calculated using the quantum model with the nuclear tunneling effect. These studies demonstrate that the charge transport within the active layer of certain OFETs is not confined solely to the interface transport. Novel research methodologies are urgently needed to facilitate the investigation of charge transport in organic semiconductor films.

Various methods exist for characterizing carrier mobility, including the time of flight (TOF) method, space-charge-limited current (SCLC) technique, and field-effect transistor method. TOF method measures mobility according to the transit time of photogenerated carrier to pass through the semiconductor [6]. The mobility measured by the TOF method is considered to be reliable, which is the bulk mobility of the film [6,25]. Using SCLC technique to evaluate the mobility of

semiconductor films, the film sample is sandwiched between two electrodes for measurement [4], revealing the bulk transport properties of the film. Field-effect transistor method is a widely used method for determining the lateral mobility of organic semiconductor films. The amount of charge within the semiconductor can be modulated by applying a voltage across the gate [6]. The mobility of the organic semiconductor can be extracted from the transfer characteristic of the organic field-effect transistor. Besides, techniques such as carrier extraction by linearly increasing voltage, double injection, and impedance spectroscopy have also been used to evaluate the mobility of organic semiconductors [4]. Nevertheless, the mobility obtained by these methods emphasizes the properties of the whole organic film or the film surface. At different positions of film, especially in the direction perpendicular to the film plane, the carrier mobility may be related to the properties of the film. These conventional methods are difficult to establish structure-property relations at different positions.

The soft plasma could incrementally and homogeneously etch the film, without damage to the materials underneath the etched surface. In the previous research, soft plasma etching was combined with light absorption spectroscopy to analyze the composition distribution of organic film along the perpendicular direction [26]. In this work, the charge transport behavior of the semiconductor film was monitored by incremental soft plasma etching. From the evolution of device characteristics, an easily accessible method for calculating the film-depth-dependent mobility was proposed. The calculation uses the electronic parameters of the transistor when the gate voltage is 0 V, and therefore requires the film containing free charge under 0 V. The suitable model films were constructed by doping or illumination. Devices based on both types of films are quite common. Doping the active layer is a commonly employed strategy to enhance transistor performance, such as improving the device's mobility [18]. Organic films with excellent

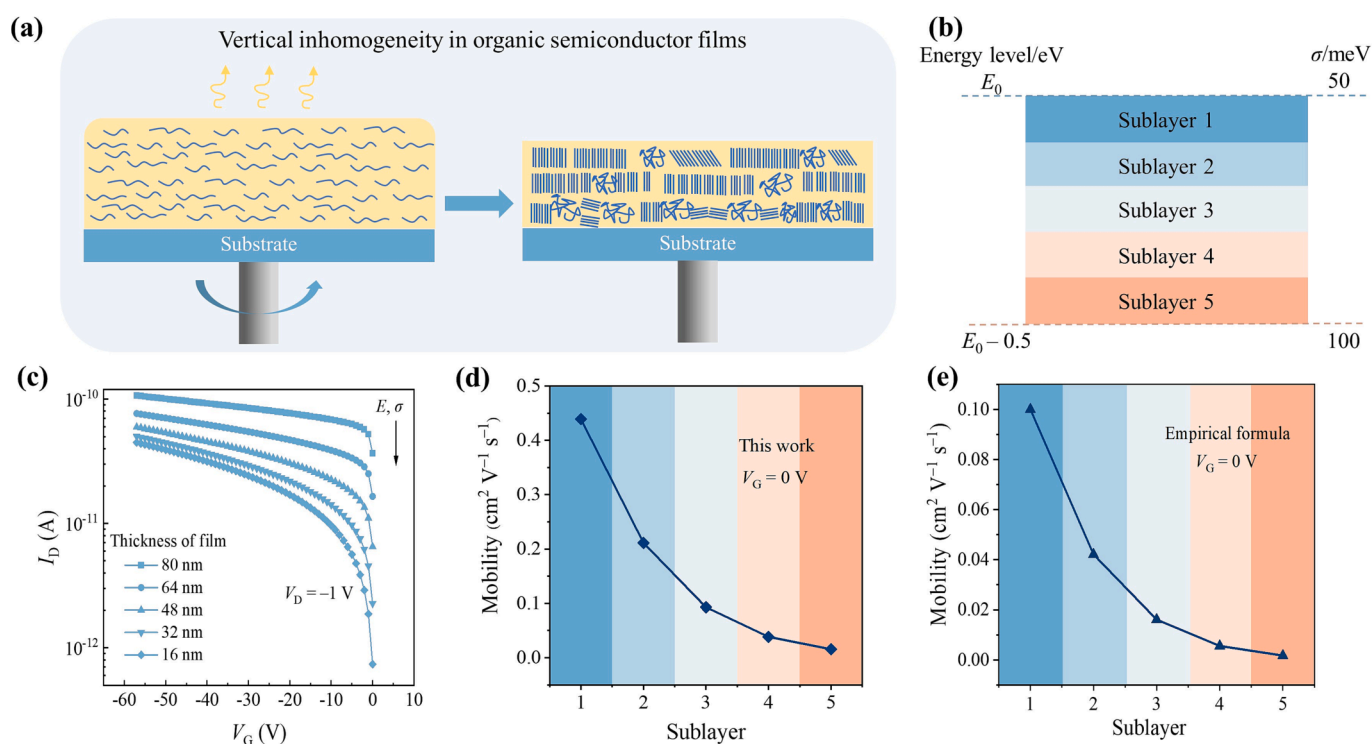


Fig. 1. Depth-dependent microstructures and charge transport behaviors of solution-processed organic semiconductor films. (a) A scheme of morphology evolution of organic semiconductor films during solvent evaporation. The film usually features vertical inhomogeneity, which could lead to film-depth-dependent charge transport characteristics. (b–e) Simulation of film-depth-dependent charge mobility of semiconductor film with vertically dispersed energetical disorders of localized states. (b) For clarity, the energy levels and disorder (σ) are arbitrarily set to change linearly from the top surface to the bottom surface of the active layer of OFETs. (c) The simulated transfer characteristics of different film thicknesses. The V_D is set to -1 V. (d) Film-depth-dependent mobility extracted from (c), which was calculated by the method proposed in this work. (e) Film-depth-dependences of mobility calculated by the empirical formula of carrier density versus carrier mobility. The detailed formulas were given in the methods section.

photoresponse have also been widely studied as active layers in optoelectronic transistors, making them suitable for applications in the field of photodetection [27]. PCDTPT (poly[4-(4,4-dihexadecyl-4H-cyclopenta[1,2-b:5,4-b']dithiophen-2-yl)-alt-[1,2,5]thiadiazolo[3,4-c]pyridine]) is a model organic semiconductor with highest occupied molecular orbital (HOMO) level of -5.14 eV [28,29]. The PCDTPT film with intrinsic charge was prepared by doping oxygen. Moreover, for the film without intrinsic charge, illumination was used to generate the free charge. IDT-BT:N2200 (indacenodithiophene copolymer with benzothiadiazole:poly{[N,N'-bis(2-octyldodecyl)-naphthalene-1,4,5,8-bis(dicarboximide)-2,6-diyl]-alt-5,5'-(2,2'-bithiophene)}) blend film exhibited obvious photoconductivity, which was also selected as the model film [30].

In this work, two conjugated model systems, doped PCDTPT film and IDT-BT:N2200 blend film were selected to investigate the film-depth-dependent electronic properties. The transfer characteristics of the transistors based on doped PCDTPT and IDT-BT:N2200 were respectively measured during incremental etching of the active layer. A series of electronic parameters were extracted from the transfer characteristics and the mobilities of the sublayers were calculated. Moreover, film-depth-dependent light absorption spectra and atomic force microscope (AFM) were used to examine the depth-dependent crystallinity and morphology. In this way, the structure-property relationship of organic semiconductor film along the perpendicular direction was successfully established. The trend of film-depth-dependent mobility was explained based on the doping degree and crystallinity of the film. Finally, the application of this method on the IDT-BT:N2200 photoelectric film was also explored. This work probably provides technical support for the further improvement of the mobility of organic semiconductor.

2. Results and discussion

2.1. Calculation of film-depth-dependent electronic properties

To investigate the electronic properties of organic semiconductor films along the vertical direction, an easily accessed approach to calculate film-depth-dependent mobility was proposed. The finite element method has been used to simulate the active layer with vertically dispersed crystallinity. As illustrated in Fig. 1b, the thickness of the semiconductor film is assumed to be 80 nm, which is tentatively divided into five sublayers. For clarity, the disorder (σ) and energy level were set to change linearly along the vertical direction. Indeed, the linear variation is not applicable to all cases in organic semiconductor films. Due to the diverse property in organic semiconductor films, it is challenging to summarize the film-depth-dependent variations with a single function. Therefore, specific problems require individual analysis. The settings here are just to simply verify the feasibility of the calculation method. The disorder reflects the crystallinity of the film, and the higher disorder represents poorer crystallinity [31]. The disorder in the figure is set according to the general range of disorder in the organic semiconductor [25]. The center of the Gaussian distribution energy level was used to describe the HOMO level [32]. The disordered phase has a deeper HOMO level [33]. When the film is used as the active layer of field-effect transistor, the mobility of the carrier in the film can be studied. When the V_G (gate voltage) approaches 0 V, the mobility is not affected by the density of gate voltage-induced charge, but determined by the intrinsic properties of the active layer. The upper sublayers of active layer are assumed to be stripped sequentially by some method (such as etching). Therefore, a method could be proposed to calculate carrier mobility in each sublayer and the calculation is based on the data when the V_G is around 0 V.

The transfer characteristics of OFETs with the remaining film as the active layer were simulated (Fig. 1c) by finite element simulation. As shown in Fig. 1c, the field-effect mobility μ of the active layer after etching is calculated as follows [34,35]:

$$\mu = \frac{L}{WC V_D} \frac{\partial I_D}{\partial V_G} \quad (1)$$

where the V_D is the source-drain voltage and I_D is the source-drain current. The C is the capacitance per unit area. The L is the channel length and W is the channel width. It is important to note that the calculation uses data at $V_G = 0$ V. The mobility obtained from formula (1) is the field-effect mobility, which is distinct from the SCLC mobility measured without the V_G applied. However, previous study has demonstrated that when V_G is around 0 V, charge accumulation and depletion primarily occur within the bulk film [36]. Consequently, the variation of I_D arises from changes in the charge density within the bulk film. At this stage, the field-effect mobility extracted from formula (1) represents the bulk mobility, which is depth-dependent.

The conductance G of the remaining active layer after etching can be calculated as follows:

$$G = \frac{I_D}{V_D} \quad (2)$$

The charge density n in the remaining active layer after etching can be calculated by the following methods:

$$\delta = ne\mu = \frac{L}{S} G \quad (3)$$

$$n = \frac{L}{Wed} \frac{G}{\mu} \quad (4)$$

δ is the conductivity of the remaining active layer after etching. e is the elementary charge 1.6×10^{-19} C. d is the thickness of the remaining active layer after etching. S is the cross-sectional area of the remaining active layer, which is equal to W multiplied by d .

The quantity of electric charge (Q) in the remaining active layer after etching is:

$$Q = neV = neSL \quad (5)$$

Combined with formula (4), the following formula can be derived:

$$Q = \frac{GL^2}{\mu} \quad (6)$$

Next, the G_i , μ_i and Q_i ($i = 1, 2, 3, 4, 5$) of the remaining active layer after etching were calculated by eq. (2), (1) and (6), respectively. The quantity of electric charge, conductance, and field-effect mobility of the active layer with the thickness of 80 nm are denoted as Q_1 , G_1 , and μ_1 , respectively. Using formulas (1), (2) and (6), μ_1 , G_1 and Q_1 can be calculated respectively. The I_D , V_D and V_G involved in the calculation process are derived from the transfer characteristics of the device with the active layer thickness of 80 nm in Fig. 1c. After sublayer 1 is etched, the above parameters in the remaining active layer (64 nm) are denoted as Q_2 , G_2 , and μ_2 , respectively. Similarly, using formulas (1), (2) and (6), μ_2 , G_2 and Q_2 can be calculated respectively. The corresponding I_D , V_D and V_G involved in the calculation process are derived from the transfer characteristics of the device with the active layer thickness of 64 nm in Fig. 1c. Taking sublayer 1 as an example, the quantity of electric charge ΔQ_1 in sublayer 1 can be obtained by subtracting the quantity of electric charge before and after the first etching:

$$\Delta Q_1 = Q_1 - Q_2 \quad (7)$$

Similarly, the conductance ΔG_1 of sublayer 1 can be calculated as follows:

$$\Delta G_1 = G_1 - G_2 \quad (8)$$

According to the following method, the mobility of carriers in sublayer 1 (μ_{s1}) is:

$$\mu_{s1} = \frac{\Delta G_1 \cdot L^2}{\Delta Q_1} \quad (9)$$

In this way, the ΔQ_i of each sublayer and the corresponding μ_{si} ($i = 1, 2, 3, 4, 5$) can be calculated in sequence. The calculation results of μ_{si} are presented in Fig. 1d. The mobilities of sublayers are revealed as a downward trend with the lower crystallinity. Based on the Poisson's equation, the charge density of each sublayer was also calculated. The film-depth-dependence of mobility was calculated by the empirical formula of carrier density versus carrier mobility [37]. The detailed formulas are given in the methods section. The resulting variation trend of mobility shown in Fig. 1e is similar to that obtained from the above method. Although there is a deviation in the mobility calculated by the two methods, it is controlled within acceptable limits. In conclusion, the variation rule of the mobility along vertical direction calculated by the method proposed in this work is reasonable and reliable. As mentioned above, the μ_{si} ($V_G = 0$ V) reflects the quality of the corresponding sublayer. This film-depth-dependent mobility probably becomes a parameter describing the property along the vertical direction of the film.

According to the above calculation, the method requires that the OFET prepared by the semiconductor film displays sufficient current at the V_G of 0 V. Therefore, the method is suitable for semiconductor films with enough charges at 0 V. For the enhancement mode p-type OFET, two-dimensional transport is shown in Fig. 2a. When $V_G < 0$ V, positive charges are induced within a few molecular layers at the interface. The positive charges are driven by V_D to form I_D . The field-induced conductance is nearly independent of the upper semiconductor [20]. For depletion mode OFET, free charges exist in the active layer when the V_G is 0 V (Fig. 2b). The charge transport is the bulk transport. Using soft plasma etching, the active layer can be stripped gradually to investigate the film-depth-dependent electronic properties (Fig. 2c). Therefore, the feasibility of the above calculation method can be verified by using the depletion OFET prepared by the required semiconductor films.

2.2. Film-depth-dependent properties of the doped PCDTPT film

For PCDTPT film, the upper surface is exposed to air and is prone to be doped by oxygen [38]. The oxygen as the electron acceptor increases the carrier density and passivates the hole traps in the PCDTPT active layer, forming depletion mode OFET [39]. Therefore, oxygen-doped PCDTPT films were studied as the model films. AFM was used for morphological characterization of the doped PCDTPT film, as shown in Fig. 3a. The morphology of oxygen-doped film is uniform before and after etching. The difference in the surface roughness is not obvious. It is probably attributed to the soft plasma etching, by which the properties of the remaining film are preserved [26]. Grazing-Incidence Wide-Angle

X-ray Scattering (GIWAXS, Fig. 3b–c) was employed to study the crystallinity of doped PCDTPT film. After etching, the peak intensity of the film decreases compared with that before etching, which can be attributed to the deterioration of crystallinity or the decrease of thickness. Further, the doped PCDTPT film was characterized using film-depth-dependent light absorption spectroscopy (Fig. 3d). With the increase of etching time, the absorbance of the film decreases gradually. The spectra (Fig. 3e) demonstrate that the peak is first red-shifted and then obviously blue-shifted along the perpendicular direction of the film. This proves that the crystallinity of doped PCDTPT film in the bottom is significantly worse than that in the middle/upper parts. And the middle part of the film exhibits excellent crystallinity [40]. Furthermore, the crystallinity of doped PCDTPT film with different thicknesses was also studied. According to Fig. S1a–b, the variation of crystallinity along the vertical direction follows a similar rule in the film prepared from a low-concentration PCDTPT solution. The peaks of the bottom are significantly blue-shifted compared with the top and middle parts. Poor crystallinity at the bottom is probably detrimental to the mobility of the location [41].

The electronic properties of bottom-gate top-contact transistor with PCDTPT film as the active layer were studied. The PCDTPT transistors were placed in air for 0 h, 1 h, and 10 h, and named as-prepared, doped (1 h), and doped (10 h) PCDTPT transistors, respectively. As Fig. 4a shows, the transfer characteristics demonstrate the p-type characteristic of the three devices. After doping in air atmosphere, I_D increases, and the threshold voltage of the OFET shifts to the positive direction. It's noted that the doping occurs within the film rather than at the interface. Due to the thin thickness of the film, the diffusion of oxygen within the semiconductor film quickly reaches equilibrium. Oxygen disperses throughout the entire film, and the extent of oxygen doping depends on the properties of the film itself [40,42,43]. Furthermore, it is apparent that the on/off ratio continuously deteriorates as the time increases from 1 h to 10 h. The off current increases significantly. The output curve of the doped (10 h) PCDTPT transistor in Fig. 4b exhibits obvious linear current regimes and saturation current regimes. The output curves show good linearity with a well-defined ohmic contact in the low V_D range. Based on the linear regime of transfer characteristics in Fig. 4a, the field-effect mobility (μ) is calculated. Fig. 4c is the relationship between μ and V_G , which was calculated from the formula (1) [34,35]. When V_G is 0 V, the μ of the doped (10 h) OFET is about $0.46 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. However, the μ of the as-prepared PCDTPT transistor can reach this value only under a more negative V_G . Previous literature has reported that a period of air exposure can passivate the hole trap and improve mobility [39]. The

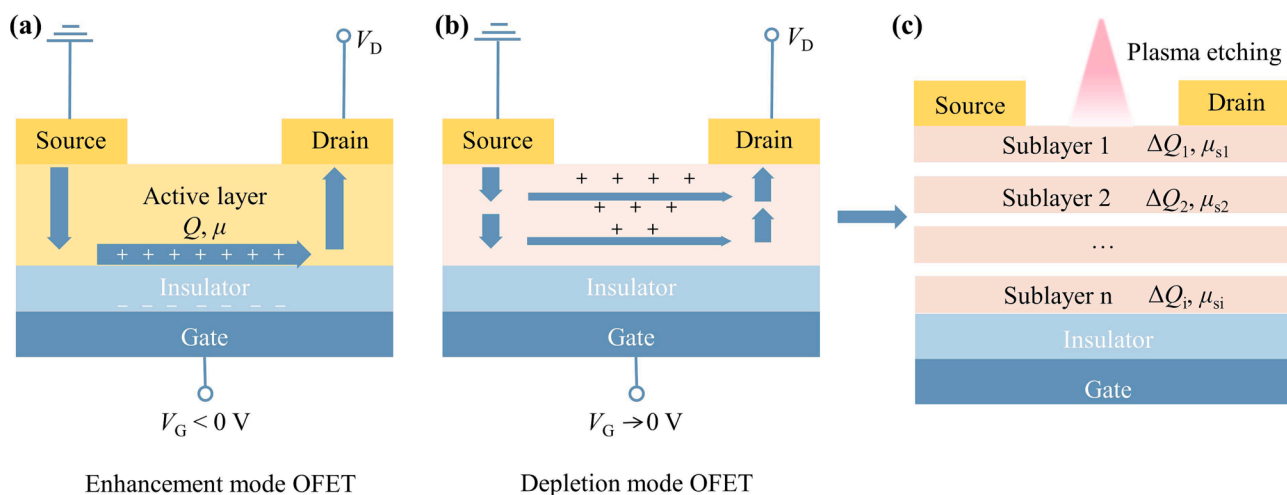


Fig. 2. Charge transport in active layer of enhancement mode OFET and bulk transport in active layer of depletion mode OFET. (a) Charge transport near the dielectric-semiconductor interface for enhancement mode OFET based on the undoped semiconductor. (b) Bulk charge distribution within depletion mode OFET based on the doped semiconductors. (c) Film-depth-dependent charge transport investigated by soft plasma etching.

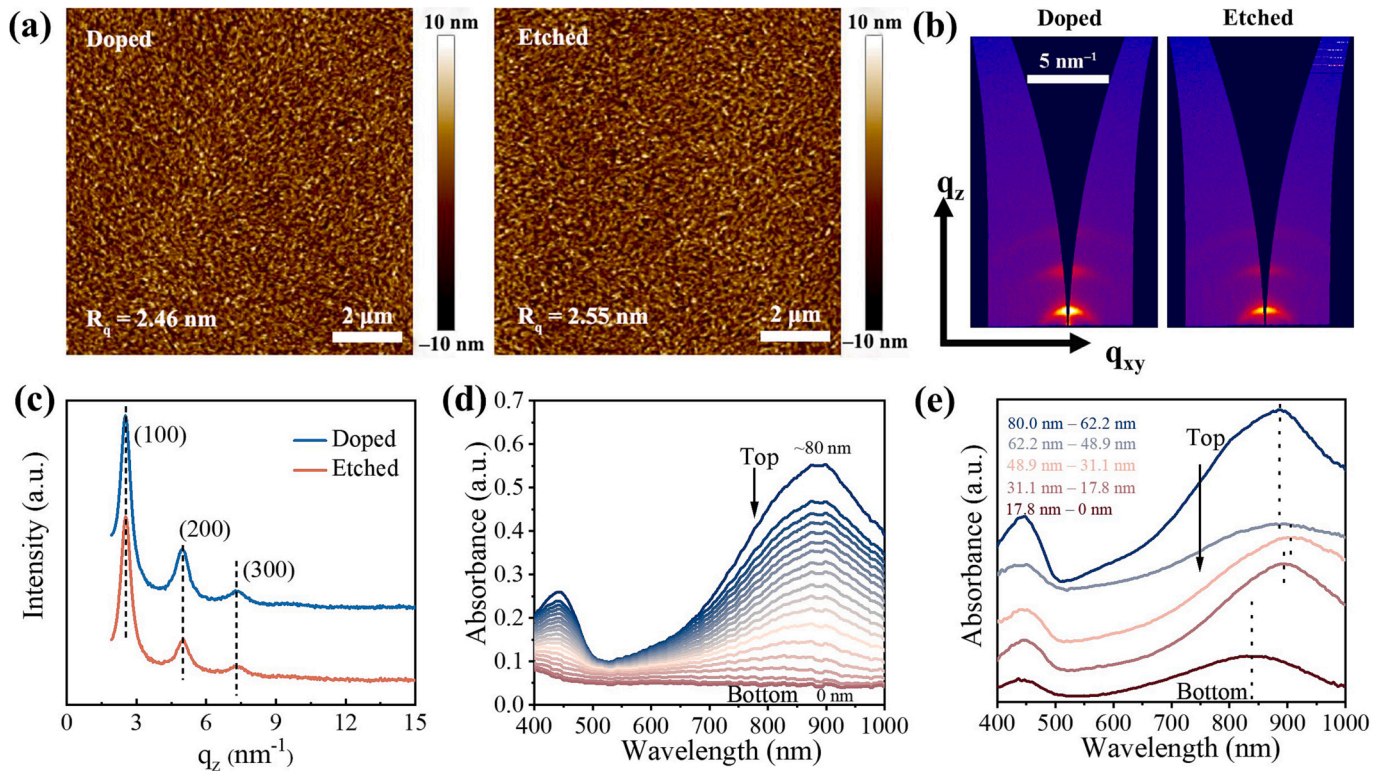


Fig. 3. Morphology of PCDTPT film before and after etching. The PCDTPT film was doped by ambient oxygen. (a) AFM height images of doped PCDTPT film before and after etching, both of which show homogenous morphology. (b) Two-dimensional GIWAXS patterns and (c) out-of-plane diffraction profiles of doped PCDTPT film before and after etching. (d) The in-situ evolution of light absorption spectra of PCDTPT film during soft plasma etching. (e) The light absorption spectra at different film-depth positions along the perpendicular direction of the doped PCDTPT film, as extracted from (d).

temperature-dependent transfer characteristics revealed the charge injection and transport mechanism of transistors. Two PCDTPT devices were fabricated and treated with insufficient doping (about 1 h) and sufficient doping (about 10 h), respectively. The transfer characteristics ($V_D = -5$ V) were measured with the temperature dropping from 30 °C to -180 °C (Fig. S2a–b). In Fig. S2c–d, the transfer characteristics were presented in the linear coordinate. The temperature-dependent linear mobility was extracted from Fig. S2 and displayed in Fig. 4d. According to the Arrhenius relation ($\mu \propto \exp(-E_A/k_B T)$), the apparent activation energy E_A of the two transistors can be calculated [44]. The E_A of the sufficiently doped transistor is 40.1 meV, which is lower than that of the insufficiently doped device (53.2 meV). It suggests that oxygen doping promotes the transport of holes.

The film-depth-dependent electronic properties of the doped PCDTPT film were calculated using doped (10 h) PCDTPT transistor. The reliability of the study on the film-depth-dependent electronic properties by means of plasma etching was illustrated in the Supplementary Note 1. As Fig. 5a (left) shows, the transfer characteristics of doped (10 h) PCDTPT transistor were tested after different etching times. With the extension of etching time, I_D decreases and the threshold voltage moves to the negative voltage direction. As the etching time increases, the transfer characteristics of the device in Fig. 5a gradually resemble those of the as-prepared device in Fig. 4a. This can be attributed to the removal of the oxygen-doped upper part, which leads to a transition from bulk transport of the charge to two-dimensional transport. The output characteristics of the device were tested after etching to investigate the effect of etching on charge transport. As shown in Fig. S5, after etching, the output characteristics still exhibit a good linear relationship and ohmic contact in the low V_D range. This demonstrates that etching did not result in a significant increase in the contact resistance of the device. According to the relationship between absorbance and film thickness [26], the film thickness also decreases linearly with etching

time (Fig. 5b). Hence, the film thickness can be roughly divided according to the etching time. As Fig. 5a (right) shows, the doped (10 h) PCDTPT active layer is artificially divided into five sublayers. The estimated film thickness is 80 nm based on the AFM measurement (Fig. S6). The thickness of the five sublayers can be calculated according to the etching time in Fig. 5a (left), which are 14.5 nm, 12.7 nm, 14.5 nm, 21.8 nm, and 16.5 nm in order. The mobility of each sublayer was calculated based on these sublayers. The thinnest sublayer is nearly 10 nm thick. Hence, the film-depth resolution for the film-depth mobility measurement is around 10 nm. The calculation process and results are presented in Fig. 5c–d. As shown in Fig. 5c, the quantity of electric charge, conductance, and carrier mobility of the active layer before etching are Q_1 , G_1 , and μ_1 , respectively. After sublayer 1 is etched, the above parameters in the remaining active layer are Q_2 , G_2 , and μ_2 , respectively. Based on the proposed method (formula (1) – (9)), the quantity of electric charge ΔQ_1 and mobility μ_{s1} in the sublayer 1 can be obtained. The quantity of electric charge per unit volume (charge density) in sublayer 1 (Q_{s1}) is:

$$Q_{s1} = \frac{\Delta Q_1}{W L d_1} \quad (10)$$

where d_1 is the thickness of sublayer 1. The quantity of electric charge per unit volume Q_{si} ($i = 1, 2, 3, 4, 5$) of each sublayer is calculated according to the thickness d_i and the charge ΔQ_i of each sublayer. As Fig. 5d shows, the Q_{s1} of the upper sublayer is more than that of the middle sublayer and lower sublayer. It's attributed to the high doping concentration of the upper sublayer compared with other sublayers. The mobility gradually decreases from the top to the bottom of the film. Specifically, the mobility of the first four sublayers is one order of magnitude higher than that of sublayer 5. Combined with the results of the absorption spectra in Fig. 3e, the crystallinity of the middle and upper parts is better than that of the bottom layer. This is the reason for

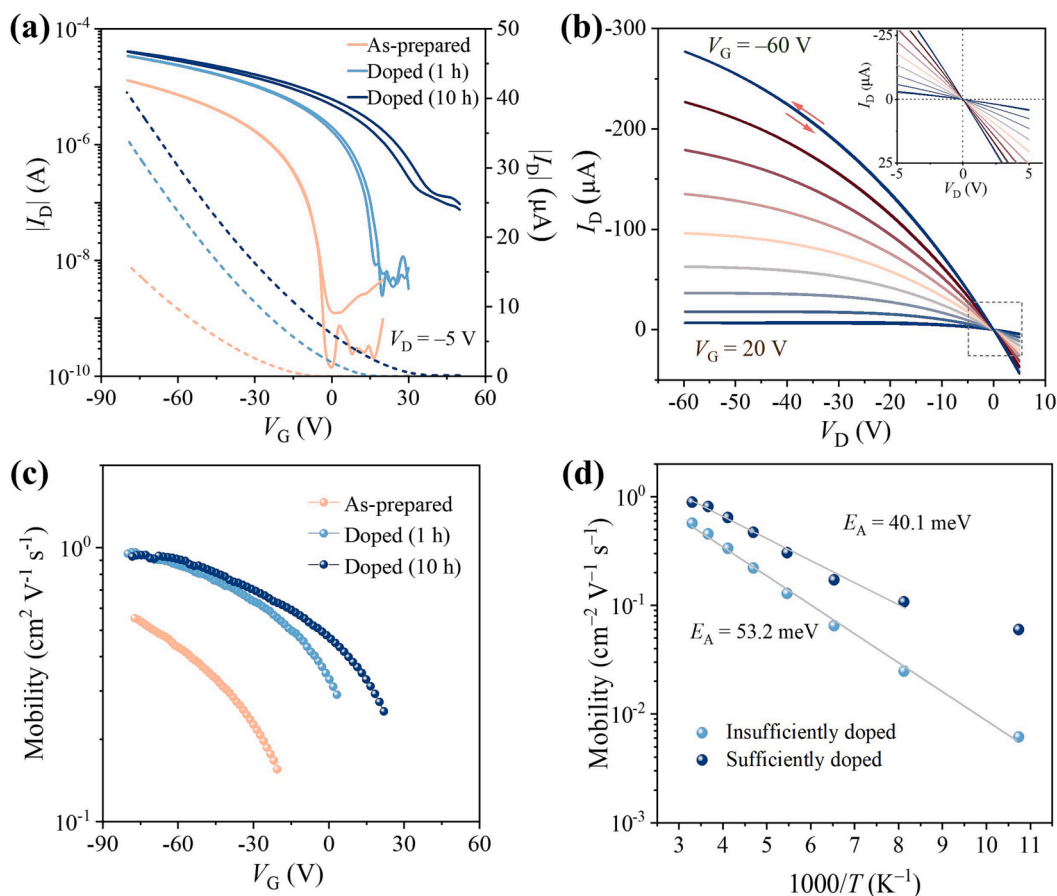


Fig. 4. Electronic properties of as-prepared and doped PCDTPT transistors. (a) Transfer characteristics of the as-prepared PCDTPT transistor and doped PCDTPT transistors. The transistors were measured in nitrogen atmosphere after exposure to air for 1 h and 10 h for doping. (b) Output characteristics of the doped (10 h) PCDTPT transistor. The inset is the output characteristic in the low V_D range, and the linear I_D - V_D characteristics show a typical ohmic contact between the electrode and semiconductor. (c) V_G -dependent field-effect mobility (linear region), extracted from the transfer characteristics in (a). (d) Temperature-dependent field-effect mobility.

the low mobility of sublayer 5. Besides, oxygen doping leads to the increase in mobility. The upper sublayer is more easily exposed to oxygen, causing the highest mobility.

The conclusion that the mobility of the top of the film is higher than that of the bottom part was verified. As illustrated in Fig. 5e, the doped PCDTPT film on the glass was floated on deionized water and picked up from the top by the Si/SiO₂ substrate. The inverted film was transferred to the Si/SiO₂ substrate and prepared into the bottom-gate/top-contact device. The substrate was not modified to avoid the cracking of the floating film caused by hydrophobic substrates. Meanwhile, the control group was also prepared by spinning PCDTPT onto the unmodified substrate. The spin-coated PCDTPT film was oxidized in air and then the electrodes were fabricated. Photographs of the devices are shown in Fig. S7. The upper and lower surfaces of the active layer prepared by the floating film method are upside-down relative to the spin-coated device. The spectra of the doped PCDTPT films on different substrates (Fig. S8) confirm the rationality of the control experiment. The transfer characteristics and mobility of devices (control group) are shown in Fig. 5f-g. The devices based on inverted film exhibit higher source-drain current, and their field-effect mobility is much higher than the field-effect mobility of the spin-coated devices (Fig. S9). This can be attributed to the fact that the upper part of the film with better electronic properties is closer to the dielectric-semiconductor interface of the device. And it acts as the main charge transport channel when the device operating. It is consistent with the findings of Hashimoto et al. regarding inverted devices, in which the superior mobility of the inverted devices is attributed to the higher molecular orientation at the polymer/air interface [45].

2.3. Film-depth-dependent properties of the IDT-BT:N2200 blend film

The feasibility of the method in the film without intrinsic charge was also probed. IDT-BT:N2200 blend film has been explored as the active layer of photoelectronic transistor in previous literature, which exhibited low source-drain current at the V_G of 0 V [30]. The IDT-BT:N2200 film was investigated by a film-depth-dependent light absorption spectrometer. The results reveal that the absorption peaks are located at 413 nm and 676 nm (Fig. S10a). Fig. S10b shows the spectra of different positions along the perpendicular direction. The red-shifted peak represents good crystallinity of that position. Compared to the spectra of other positions, the peak of the film is significantly blue-shifted at the bottom, indicating the deteriorative crystallinity. This probably results in poor mobility of carriers at the bottom of the film.

To study the film-depth-dependences of mobility of IDT-BT:N2200 film, transistor with bottom-gate top-contact structure (Fig. 6a) was fabricated. Red light with a wavelength of 625 nm was used as the light source to get the high source-drain current at V_G of 0 V. The output characteristics of IDT-BT:N2200 transistor (Fig. 6b) show approximately linearity over the low V_G range, confirming that the influence of contact can be eliminated. The transfer characteristics in Fig. 6c exhibits obvious photoresponsivity under illumination condition. Further, the active layer of IDT-BT:N2200 transistor was treated with soft oxygen plasma, and the transfer characteristics after each etching treatment were measured (Fig. 6d). With the increase in etching time, the I_D decreased gradually. After etching for 330 s, only a very low I_D signal could be detected (this curve has not been plotted in Fig. 6d). The active layer can

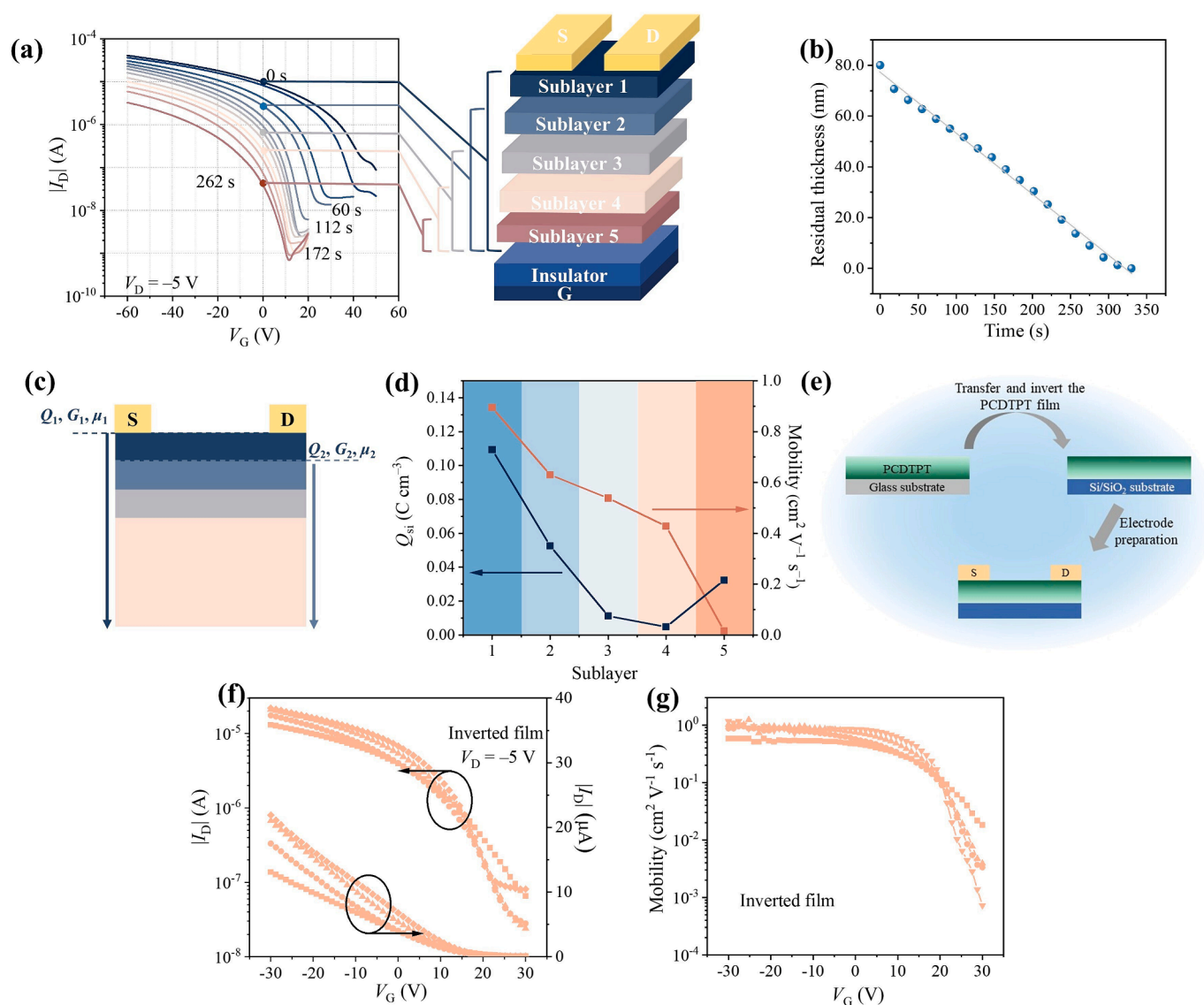


Fig. 5. Film-depth-dependent charge density and mobility of PCDTPT film. (a) Transfer characteristics (left) of the doped (10 h) PCDTPT transistor during etching for different times. Before etching, the source and drain electrodes have been deposited. The diagram (right) displays corresponding sublayers of the marked transfer characteristics. (b) The thickness evolution of the active layer of the doped PCDTPT transistor during etching is estimated from the absorption spectra in Fig. 3d. (c) Schematic diagram of the calculation of film-depth-dependent mobility. (d) The film-depth-dependent charge density and film-depth-dependent mobility, as extracted from (a). (e) The fabrication process of PCDTPT transistor with inverted film. The doped PCDTPT film was floated on deionized water and picked up from the top with an unmodified Si/SiO₂ substrate. (f) Transfer characteristics of the PCDTPT transistors (multiple parallel samples) with inverted film as the active layer. (g) V_G -dependent field-effect mobility, as extracted from the transfer characteristics in (f).

be roughly considered to be completely etched.

It is inconvenient to simultaneously adjust the light intensity during the measurement. Hence, as shown in Fig. 6e (left), the light intensity was kept constant during the measurement. To simulate the real situation of light irradiation from the upper surface of the active layer, the absorption of light by the etched sublayer needs to be taken into account. That is, I_D extracted from Fig. 6d needs to be calibrated according to Fig. 6e (right). The detailed current correction process is presented in Supplementary Note 2. Based on the corrected current, the film-depth-dependent mobilities obtained using the formula (1) – (9) are drawn in Fig. 6f. The average value of mobility fluctuates in the range of 0.2–0.3 $cm^2\ V^{-1}\ s^{-1}$ in the first three sublayers. Compared with the PCDTPT system, the first three sublayers of the IDT-BT:N2200 film exhibit similar mobility. It is speculated that this may be related to the lower I_D of the IDT-BT:N2200 system compared to the PCDTPT system when V_G is at 0 V. This results in insignificant differences between the sublayers of IDT-BT:N2200 film. The mobility is suddenly reduced at the

bottom of the film. This is related to the poor crystallinity of the bottom.

The above experiments demonstrate that the method to calculate film-depth-dependent mobility is feasible to study film with significant photoresponsivity. However, the quantum efficiency of the active layer and the interference of the light are not taken into account in the calculation process of this work. Therefore, the results can only be used to roughly evaluate the quality of the film in the vertical direction. Further research is needed to refine the method.

3. Conclusion

In summary, an easily accessible method is proposed to measure the vertical distribution of charge density and mobility in organic semiconductor films. The film-depth resolution for the film-depth-dependent mobility measurement is around 10 nm. Soft plasma incrementally etched the film without damage to the underneath materials, while the transfer characteristics of the semiconductor layer were in situ

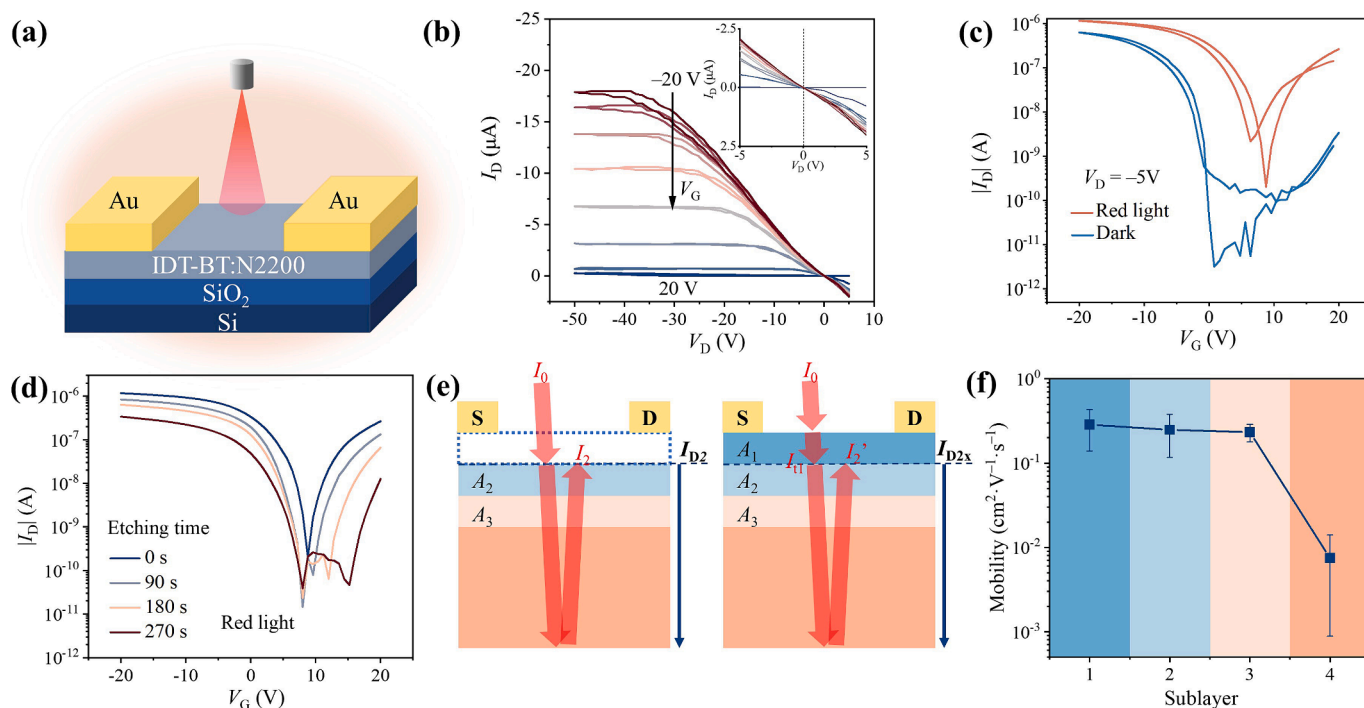


Fig. 6. The film-depth-dependences of mobility of the film without intrinsic charges. Here, IDT-BT:N2200 blend film was employed as an example, which was measured under illumination. (a) The configuration of IDT-BT:N2200 transistor studied in this work. The light source used in the measurement is red light with a wavelength of 625 nm ($\sim 17 \text{ mW cm}^{-2}$). (b) Output characteristics of IDT-BT:N2200 transistor. (c) Transfer characteristics of IDT-BT:N2200 transistors under illumination and dark conditions. The V_D was set to -5 V . (d) Transfer characteristics of IDT-BT:N2200 transistor after etching for different time. (e) The schematic light propagation in the OFET during etching. (f) The film-depth-dependences of mobility with error bars, which was calculated according to the data in (d) and several duplicate devices.

measured. Charge density, mobility and their dependences on film depth were extracted from the evolution of transfer characteristics during soft plasma etching. The doping degree and film-depth-dependent crystallinity were confirmed as factors affecting the film-depth-dependent mobility of doped PCDTPT film. It is found that for PCDTPT, the mobility at the top surface is one order higher than that at the buried region of the transistor. Moreover, the method could also be applied to the material without sufficient intrinsic charges, for example, the IDT-BT:N2200 photoactive film under light illumination. The film-depth-dependent mobility measurement could be applicable for characterizing many thin-film electronic devices. This work provides insights into anisotropic charge transport in organic photoelectronic films and technical support for the further enhancement of the performance of organic semiconductors.

4. Methods

Materials. All the chemical reagents were commercially purchased and used without further purification. PCDTPT used in this work was obtained from 1-Material Inc. IDT-BT and N2200 were purchased from Organtec Ltd.

Device Fabrication. In the ultrasonic cleaning machine, the Si/SiO₂ substrates were cleaned for 10 min by deionized water, acetone, and isopropyl alcohol successively. Subsequently, the substrates were dried with nitrogen and treated with oxygen plasma to clean the surface. After that, the Si/SiO₂ substrates were modified with octyltrichlorosilane (OTS) at 85°C for 2 h, and the solvent was toluene. To prepare the active-layer solution, the PCDTPT was dissolved in chloroform (CF) and the concentration of this polymer solution was 10 mg mL^{-1} . Thinner PCDTPT film was prepared from a solution of 5 mg mL^{-1} . The PCDTPT layer was fabricated through the spin coating (2000 rpm). And then these substrates with active layers were annealed at 150°C for 30 min. The source and drain electrodes (50 nm) were fabricated by evaporating

gold. The length (L) and width (W) of the device channel are $300 \mu\text{m}$ and 3 mm , respectively. The thickness of SiO₂ is 300 nm . The calculation of mobility utilized a capacitance value of 11.5 nF cm^{-2} . The doped PCDTPT transistor was obtained by placing the PCDTPT transistor in an ambient atmosphere.

Preparation of the floating film: poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS) was spin coated on the glass substrate, and then PCDTPT film was fabricated through the spin coating. After annealing, the film was exposed to air for 10 h. Being soaked in deionized water, the doped PCDTPT film was floated as the PEDOT:PSS dissolved.

IDT-BT:N2200 transistor used Si/SiO₂ substrate with the same treatment as the PCDTPT transistor. In the solution of the active layer, IDT-BT and N2200 were dissolved in CF with a total concentration of 10 mg mL^{-1} . The mass of N2200 is 5% of the total mass of IDT-BT and N2200. The active layer was prepared by a spin coating process, and its parameters were 3000 rpm and 60 s . After that, the substrate and active layer were annealed at 100°C for 1 h. The source and drain were prepared in the same way as the PCDTPT transistor. The channel length and width are $100 \mu\text{m}$ and 3 mm , respectively.

Characterization. The performances of the PCDTPT transistors in Fig. 4 and Fig. 5 were tested using the Agilent Keysight B2902A Precision Source/Measure Unit. The data in Fig. 4 was measured inside a nitrogen-filled glovebox. The measurements for Fig. 5 were conducted under ambient air conditions. The performance of the IDT-BT:N2200 transistor was measured in an ambient atmosphere. To calculate the film-depth-dependent mobility, the transistors were first treated by a plasma-etching machine, and then the transfer characteristics were measured. GIWAXS test was employed to study the crystallinity of the film at P03 beamline of Deutsches Elektronen Synchrotron (DESY), Germany. The X-ray wavelength used was $1.05 \text{ \AA}$, and the incident angle (α) was 0.15° . A LAMBDA 4.5 M detector was utilized to capture the two-dimensional GIWAXS patterns, while the sample-to-detector

distance was calibrated using a cerium dioxide standard sample. AFM (INNOVA) was used to investigate the surface morphology. The XPS (X-ray photoelectron spectroscopy) was carried out using Thermo Fisher ESCALAB Xi⁺. Film-depth-dependent light absorption spectrometer was applied to investigate the composition and crystallinity of PCDTPT and IDT-BT:N2200 films, and the measurement method is available elsewhere [46]. For measurements of film-depth-dependent light absorption spectra, doped PCDTPT and IDT-BT:N2200 films were prepared on glass substrates treated with OTS. The preparation process is the same as that of the active layer of the transistor mentioned above. The transistors were etched by oxygen plasma etching at the pressure of less than 20 Pa with an individual plasma setup (Diener). The film on the glass was treated at a pressure of about 20 Pa, a power of 20 W with the film-depth-dependent light absorption spectrometer. The film-depth-dependent light absorption spectrometer is a homemade device, which integrates etching and spectra analysis functions to measure in situ the evolution of spectra in depth direction. It can be used to analyze the depth-dependent composition and crystalline evolution of the films [47]. Both the individual plasma setup and film-depth-dependent light absorption spectrometer operate based on the mechanism of low-pressure plasma etching. Although the etching rates are different due to the variations in equipment, this does not affect the conclusions of the manuscript since the devices operate based on the same etching principle.

Calculation. The vertical direction of the active layer was simulated by the finite element method, obtaining the transfer characteristics during etching. According to the transfer characteristic, the method to calculate film-depth-dependent mobility was used to solve the mobility distribution along the perpendicular direction. The threshold voltage of the simulated transfer characteristics in Fig. 1c was artificially set. First, the trap density of the polymer semiconductor was defined. And the induced charge density at different V_G was compared with the trap density. Furthermore, for the algorithm to run, a background charge density was defined. If the induced charge density is equal to the trap density, then the trap is just filled. The corresponding V_G is the threshold voltage. Thicker films accommodate more hole charges and more positive threshold voltage is required to deplete the charges. In the calculation process, the following approximate treatment was made: the channel current consists only of the drift current. Gradual channel approximation was applied to the device. Besides, the charge density of each sublayer was also obtained by solving Poisson's Equation with input of Fermi-Dirac distribution. Then the mobility of the sublayer was calculated by empirical formula [37]:

$$\mu(T, n) = \mu_0(T) \exp \left[\frac{1}{2} (\hat{\sigma}^2 - \hat{\sigma}) (2na^3)^{\hat{\sigma}} \right]$$

$$\hat{\sigma} \equiv 2 \frac{\ln(\hat{\sigma}^2 - \hat{\sigma}) - \ln(\ln(4))}{\hat{\sigma}^2}, \hat{\sigma} \equiv \frac{\sigma}{k_B T} \quad (11)$$

where $\mu_0(T)$ is a prefactor mobility. a is the lattice constant. σ is the width of Gaussian distribution. k_B is the Boltzmann constant, and T is temperature. The calculation method has been reported in our group's previous research work [36].

Author contributions

The manuscript was written through the contributions of all authors. Guanghao Lu conceived and supervised the project. Peng Wei and Qingyu Wang designed and conducted the experiments. Zongze Qin performed the finite element simulation. Yixin Ran designed the experiment on the films under illumination. Peng Zhang conducted the GIWAXS test. All authors participated in the discussion and commented on the manuscript. All authors have approved the final version of the manuscript. Portions of this research were carried out at the light source PETRA III at DESY. DESY is a member of the Helmholtz Association

(HGF).

CRediT authorship contribution statement

Qingyu Wang: Writing – original draft, Investigation. **Peng Wei:** Investigation. **Zongze Qin:** Investigation. **Yixin Ran:** Investigation. **Xin Wang:** Writing – review & editing. **Peng Zhang:** Investigation. **Guanghao Lu:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.150202>.

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